David Kimanthi
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Career Summary

Results-driven Software Engineer with 5+ years of experience building scalable backend systems and cloud applications. Proficient in Python, SQL, and AWS with strong focus on code quality, testing, and CI/CD practices. Proven track record developing APIs, microservices, and distributed systems using modern frameworks and containerization technologies. AWS Certified with experience in system design, performance optimization, and production reliability.

Work Experience

- **IDinsight** Nairobi, Kenya
  Software Engineer - Data Systems *Aug 2022 – Present*
  - **Built and operated Python/FastAPI microservices on AWS** to expose real-time analytics and program-management APIs used across 8 semi-urban social programs
  - **Translated complex stakeholder requirements into scalable backend architectures (domain modeling, API versioning, data access layers), aligning business needs with technical implementation and rapid feedback loops.**
  - **Designed RESTful API contracts with Pydantic schemas and SQLAlchemy models**
  - **Introduced testing discipline** (pytest, factory-based fixtures) and code review standards; mentored junior engineers, improving team productivity and code quality.
  - **Core Technologies:** AWS Cloud Services | Python | Terraform | FastAPI | PySpark | PostgreSQL | Docker | Power BI Integration | Airbyte

- **Hence Technologies** Kigali, Rwanda
  Data Engineer *Sep 2021 – Jul 2022*
  - **Collaborated with product teams** to design data ontologies aligned with customer workflows, **reducing data integration complexity and accelerating feature development cycles**
  - **Built automated PySpark data pipelines** processing legal datasets, **eliminating 15+ hours of manual processing weekly and enabling real-time product updates**
  - **Developed ML pipeline** for legal named entity recognition using SpaCy, **enriching product database with more relevant content and improving user search accuracy**
  - **Established data engineering standards** and documentation, **reducing onboarding time for new engineers by 50% and improving code maintainability**
  - **Core Technologies:** Python | PySpark | SpaCy | NLP | Pandas | Spark SQL | Machine Learning Operations

- **Safaricom PLC** Nairobi, Kenya
  Software Engineer in Test *Jul 2019 – Aug 2020*
  - **Developed ReactJS reporting dashboard** for test automation results, **reducing test result delivery time by 60% and providing real-time visibility to over 20 developers across 5 teams**
  - **Built comprehensive API automation framework** (SOAP/REST) using pytest, **achieving 50% reduction in manual testing workload and eliminating testing bottlenecks for critical services**
  - **Integrated Python automation framework with Jenkins CI/CD pipelines**, **ensuring micro-service uptime through automated daily testing and comprehensive failure reporting**
  - **Core Technologies:** JavaScript | Python | PyTest | Jenkins | Docker | Selenium | ReactJS | Jira

- **PricewaterhouseCoopers (PwC)** Nairobi, Kenya
  Software Application Developer *Jul 2016 – April 2019*
  - **Led enterprise data migration** from legacy on-premise systems to Salesforce cloud, **successfully migrating 100% of critical client data with zero downtime**
  - **Built ETL packages** using SQL Server Integration Services (SSIS), **streamlining cross-system data exchange between on-premise and cloud platforms**
  - **Executed comprehensive User Acceptance Testing** across multiple project implementations, **ensuring compliance with business requirements**
  - **Core Technologies:** SQL Server | T-SQL | SSIS | JavaScript | C-Sharp | .Net Core

Education

- **Carnegie Mellon University** Pittsburgh, US
  Master's Degree in Information Technology *2020 – 2022*
- **University of Nairobi** Nairobi, Kenya
  Bachelor of Science in Computer Science *Sep 2012 – May 2016*

Certifications

AWS Certified Data Engineer – Associate

AWS Certified Solutions Architect – Associate